Solutions – Midterm Exam 2

ECON 320 – Introduction to Mathematical Economics

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Instructor: Muhebullah Karimzada

Problem 1 (12 points)

A firm's profit: $\pi(Q) = 120Q - 6Q^2 - 180$.

(a) First-order condition and critical point (4 pts).

$$\pi'(Q) = 120 - 12Q \quad \Rightarrow \quad \pi'(Q) = 0 \Rightarrow 120 - 12Q = 0 \Rightarrow Q^\star = 10.$$

(b) Second derivative and classification (3 pts).

$$\pi''(Q) = -12 < 0 \quad \Rightarrow \quad \text{strictly concave} \Rightarrow Q^\star = 10 \text{ is a (global) maximum.}$$

(c) Maximum profit (3 pts).

$$\pi^\star = \pi(10) = 120(10) - 6(10)^2 - 180 = 1200 - 600 - 180 = 420.$$

(d) Interpretation (2 pts). $Q^\star$ is the profit-maximizing output. Concavity ($\pi'' < 0$) means marginal profit is decreasing; the unique stationary point is the global maximizer.

Problem 2 (12 points)

Population $P(t) = 200e^{0.03t}$, technology $A(t) = e^{0.02t}$, output $Y(t) = A(t)\,P(t)^{0.5}$.

(a) Instantaneous growth rate of $Y(t)$ via log-differentiation (4 pts).

$$\ln Y(t) = \ln A(t) + \tfrac{1}{2}\ln P(t) = 0.02t + \tfrac{1}{2}\Big(\ln 200 + 0.03t\Big) = \tfrac{1}{2}\ln 200 + (0.02 + 0.015)t.$$

Therefore

$$\frac{d}{dt}\ln Y(t) = 0.035 \quad \Rightarrow \quad \frac{1}{Y(t)}\frac{dY}{dt} = 0.035.$$

So Y grows at 3.5% per unit time; equivalently $\dfrac{dY}{dt} = 0.035\,Y$.

(b) Compute $Y(10)$ (3 pts).

$$Y(t) = \sqrt{200}\, e^{0.035t} \quad \Rightarrow \quad Y(10) = \sqrt{200}\, e^{0.35}.$$

Using $\sqrt{200} \approx 14.1421$ and $e^{0.35} \approx 1.4191$,

$$Y(10) \approx 14.1421 \times 1.4191 \approx 20.07.$$

(c) Higher population growth: $0.03 \to 0.04$ (3 pts).

$$\frac{1}{Y}\frac{dY}{dt} = \frac{d}{dt}\ln Y = 0.02 + \tfrac{1}{2}(0.04) = 0.02 + 0.02 = 0.04.$$

Thus the new growth rate is 4%.

(d) Interpretation (2 pts). Output growth equals TFP growth (2%) plus labor/population growth weighted by its elasticity (0.5). A 1 percentage-point increase in population growth raises the output growth rate by 0.5 percentage points, showing the relative roles of technology and population.

Problem 3 (13 points)

$f(x, y) = 12x + 8y - x^2 - y^2 - xy.$

(a) First-order conditions and stationary point(s) (4 pts).

$$f_x = 12 - 2x - y, \qquad f_y = 8 - 2y - x.$$

Solve $f_x = f_y = 0$:

$$\begin{cases} 12 - 2x - y = 0 \\ 8 - 2y - x = 0 \end{cases} \Rightarrow \begin{cases} y = 12 - 2x \\ 8 - 2(12 - 2x) - x = 0 \end{cases} \Rightarrow -16 + 3x = 0 \Rightarrow x^\star = \frac{16}{3}, \quad y^\star = 12 - 2 \cdot \frac{16}{3} = \frac{4}{3}.$$

(b) Hessian and determinant (4 pts).

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}, \qquad \det(H) = (-2)(-2) - (-1)^2 = 4 - 1 = 3 > 0.$$

(c) Classification (3 pts). Since $f_{xx} = -2 < 0$ and $\det(H) > 0$, H is negative definite. Therefore $(x^\star, y^\star)$ is a strict local maximum. Because H is constant and negative definite everywhere, it is also the global maximizer.

(d) Economic interpretation (2 pts). Negative diagonals indicate diminishing returns to each input; $f_{xy} = -1 < 0$ shows inputs are gross substitutes (raising one reduces the marginal product of the other). Global concavity implies a unique optimum.

Problem 4 (13 points)

Maximize $U(x, y) = x^{0.5}y^{0.5}$ subject to $2x + y = 100$.

(a) Lagrangian and FOCs (4 pts).

$$\mathcal{L}(x, y, \lambda) = x^{1/2}y^{1/2} - \lambda(2x + y - 100).$$

$$\frac{\partial \mathcal{L}}{\partial x} = \tfrac{1}{2}x^{-1/2}y^{1/2} - 2\lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = \tfrac{1}{2}x^{1/2}y^{-1/2} - \lambda = 0, \quad 2x + y - 100 = 0.$$

From the first two FOCs, equate:

$$\tfrac{1}{2}x^{-1/2}y^{1/2} = 2\lambda, \qquad x^{1/2}y^{-1/2} = \lambda \;\Rightarrow\; \frac{\tfrac{1}{2}x^{-1/2}y^{1/2}}{x^{1/2}y^{-1/2}} = 1 \;\Rightarrow\; \tfrac{1}{2} \cdot \frac{y}{x} = 1 \Rightarrow \frac{y}{x} = 2.$$

Thus $y = 2x$. Budget: $2x + y = 4x = 100 \Rightarrow x^\star = 25,\ y^\star = 50$. Multiplier from $\lambda = \frac{1}{2}x^{1/2}y^{-1/2}$:

$$\lambda^\star = \frac{1}{2}\sqrt{\frac{x^\star}{y^\star}} = \frac{1}{2}\sqrt{\frac{25}{50}} = \frac{1}{2\sqrt{2}} \approx 0.3536.$$

(c) Bordered Hessian check (2 pts). Second derivatives:

$$U_{xx} = -\tfrac{1}{4}x^{-3/2}y^{1/2}, \quad U_{yy} = -\tfrac{1}{4}x^{1/2}y^{-3/2}, \quad U_{xy} = \tfrac{1}{4}x^{-1/2}y^{-1/2}.$$

Constraint $g(x, y) = 2x + y - 100$ gives $g_x = 2,\ g_y = 1$. The bordered Hessian

$$H_B = \begin{bmatrix} 0 & 2 & 1 \\ 2 & U_{xx} & U_{xy} \\ 1 & U_{xy} & U_{yy} \end{bmatrix}_{(x^\star, y^\star)}$$

has $\det(H_B) > 0$ at $(25, 50)$, confirming a local maximum. Alternatively, note U is strictly concave on $\mathbb{R}^2_{++}$ and the constraint is linear, implying a unique global maximum.

(d) Interpreting $\lambda^\star$ and envelope estimate (2 pts). With budget $2x + y = I$ and $I = 100$, the envelope theorem gives $\partial U^\star/\partial I = \lambda^\star$, the marginal utility of income. For $\Delta I = 5$,

$$\Delta U \approx \lambda^\star \Delta I = \frac{1}{2\sqrt{2}} \times 5 \approx 1.7678.$$